Mathematical English–Hindi Lexicon & Teaching Handbook

Comprehensive 250-Term Reference for Mathematics Classroom Discourse, Pedagogy & Formal Academic Writing

PART 1: FOUNDATIONAL MATHEMATICAL NOUNS (ITEMS 1–25)

Nouns: Set A

#	Term (Noun)	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
1	Equation	समीकरण (Sameekaran)	A mathematical statement asserting equality.	*"Balance both sides of the linear equation carefully."*
2	Variable	चर राशि (Char Rashi)	A symbol representing an unspecified quantity.	*"Let variable x denote the unknown integer."*
3	Coefficient	गुणांक (Gunaank)	A numerical factor multiplying a variable.	*"The coefficient of x^2 in this quadratic term is 5."*
4	Numerator	अंश (Ansh)	The top number in a common fraction.	*"Multiply both numerators before simplifying the fraction."*
5	Denominator	हर (Har)	The bottom number denoting the divisor.	*"A fraction is undefined if its denominator equals zero."*
6	Quotient	भागफल (Bhagphal)	The result obtained by dividing quantities.	*"Dividing thirty-six by four yields a quotient of nine."*
7	Remainder	शेषफल (Sheshphal)	The integer leftover after division.	*"Seventeen divided by five leaves a remainder of two."*
8	Polynomial	बहुपद (Bahupad)	An algebraic sum of multiple monomial terms.	*"This cubic polynomial possesses at most three real roots."*
9	Exponent	घातांक (Ghataank)	Power indicating repeated multiplication.	*"Apply exponential laws to add exponents of like bases."*
10	Integer	पूर्णांक (Poornaank)	A whole number (positive, negative, or zero).	*"Verify whether the resulting solution is an integer."*
11	Fraction	भिन्न (Bhinn)	A quotient showing parts of a whole.	*"Express the decimal 0.75 as an irreducible fraction."*
12	Theorem	प्रमेय (Pramey)	A proposition proven via deductive reasoning.	*"We apply Pythagoras' theorem to find the hypotenuse."*
13	Axiom / Postulate	स्वयंसिद्ध (Swayamsiddha)	A self-evident foundational premise.	*"Euclidean geometry accepts this statement as an axiom."*
14	Hypotenuse	कर्ण (Karn)	Longest side facing the 90° angle.	*"Calculate the hypotenuse using the two perpendicular legs."*
15	Perimeter	परिमाप (Parimaap)	Continuous line forming boundary of a figure.	*"Add all four boundary lengths to compute the perimeter."*
16	Circumference	परिधि (Paridhi)	The enclosing boundary of a circular figure.	*"The circumference equals two pi multiplied by the radius."*
17	Radius	त्रिज्या (Trijya)	Segment from center to boundary circle point.	*"Double the circle's radius to determine its diameter."*
18	Diameter	व्यास (Vyaas)	Straight line segment passing through center.	*"The chord of maximal length across any circle is its diameter."*
19	Coordinate	निर्देशांक (Nirdeshaank)	Ordered pair determining point position.	*"Plot the point corresponding to coordinate (3, -4)."*
20	Probability	प्रायिकता / संभावना (Prayikta)	Quantified likelihood of an event occurring.	*"The probability of rolling a composite number is one-half."*
21	Matrix	आव्यूह (Aavyooh)	Rectangular array of quantities in rows/columns.	*"Check if the square matrix possesses a non-zero determinant."*
22	Determinant	सारणिक (Saarnik)	Scalar value computed from square matrix entries.	*"If the determinant vanishes, the matrix has no inverse."*
23	Derivative	अवकलज (Avkalaj)	Instantaneous rate of change of a function.	*"Compute the first derivative to determine turning points."*
24	Integral	समाकल (Samakal)	Continuous accumulation / area function	*"Evaluate the definite integral over the interval $[0, \pi]$."*

PART 1: MATHEMATICAL NOUNS CONTINUED (ITEMS 26–50)

Nouns: Set B

#	Term (Noun)	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
26	**Vector**	**सदिश (Sadish)**	Quantity possessing magnitude and direction.	*"Resolve the velocity vector into orthogonal components."*
27	**Scalar**	**अदिश (Adish)**	Quantity described purely by magnitude.	*"Mass and kinetic energy are purely scalar quantities."*
28	**Logarithm**	**लघुगणक (Laghuganak)**	Exponent required to produce a given number.	*"Transform the product into a sum by taking natural logarithms."*
29	**Sequence**	**अनुक्रम (Anukram)**	Ordered succession of mathematical terms.	*"Establish the recurrence relation for the Fibonacci sequence."*
30	**Series**	**श्रेणी (Shreni)**	Sum of elements belonging to a sequence.	*"Determine whether this infinite geometric series converges."*
31	**Parabola**	**परवलय (Parvalay)**	Conic section formed by locus of equidistant points.	*"The vertex represents the extreme value of this parabola."*
32	**Ellipse**	**दीर्घवृत्त (Deerghvrit)**	Closed curve with sum of focal distances constant.	*"Planetary orbits trace an ellipse with the sun at one focus."*
33	**Hyperbola**	**अतिपरवलय (Atiparvalay)**	Open curve with constant difference of focal radii.	*"The graph of $xy = c$ forms a rectangular hyperbola."*
34	**Mean**	**माध्य / औसत (Maadhya)**	Arithmetic average value of a data set.	*"Divide the aggregate total by sample size to get the mean."*
35	**Median**	**माध्यिका (Maadhyika)**	The middle value in an ordered statistical set.	*"Unlike the mean, the median is resilient against outliers."*
36	**Mode**	**बहुलक (Bahulak)**	Value appearing with greatest statistical frequency.	*"The distribution is bimodal since two peaks share equal mode."*
37	**Variance**	**प्रसरण (Prasaran)**	Expectation of squared deviations from mean.	*"A higher variance implies broader spread across scores."*
38	**Standard Deviation**	**मानक विचलन (Manak Vichlan)**	Square root of variance showing dispersion.	*"Approximately 68% of data falls within one standard deviation."*
39	**Tangent**	**स्पर्शरेखा (Sparshrekha)**	Straight line touching a curve at exactly one point.	*"The slope of the tangent represents the derivative at that point."*
40	**Secant**	**छेदक रेखा (Chhedak Rekha)**	Line cutting across two points on a curve.	*"As endpoints converge, the secant line becomes a tangent."*
41	**Origin**	**मूल बिंदु (Mool Bindu)**	Point of intersection (0,0) of coordinate axes.	*"Notice the line passes directly through the origin."*
42	**Quadrant**	**चतुर्थांश (Chaturthaansh)**	One of four divisions on a Cartesian plane.	*"Both coordinates remain strictly positive in Quadrant I."*
43	**Prism**	**प्रिज्म (Prism)**	Polyhedron with identical parallel polygonal ends.	*"Find the total surface area of this triangular prism."*
44	**Polyhedron**	**बहुफलक (Bahufalak)**	Three-dimensional solid with flat polygonal faces.	*"A cube is a regular polyhedron possessing six faces."*
45	**Subset**	**उपसमुच्चय (Upsamucchay)**	Set whose elements are contained within another.	*"The set of natural numbers forms a subset of integers."*
46	**Intersection**	**सर्वनिष्ठ (Sarvanisht)**	Elements common to two or more given sets.	*"The intersection of sets A and B contains only odd primes."*
47	**Union**	**सम्मेलन / संघ (Sangh)**	Collection of all elements in combined sets.	*"Take the union of both intervals to write the full domain."*
48	**Diagonal**	**विकर्ण (Vikarn)**	Segment linking non-adjacent polygon vertices.	*"The diagonals of a rhombus bisect each other perpendicularly."*
49	**Sector**	**त्रिज्यखंड (Trijyakhand)**	Portion of a circle bounded by two radii and arc.	*"Calculate the area of the circular sector subtending 60 degrees."*
50	**Trapezoid**	**समलम्ब (Samlamb)**	Quadrilateral with at least one pair of parallel sides.	*"Average the parallel bases to compute trapezoid area."*

PART 2: CORE MATHEMATICAL ADJECTIVES (ITEMS 51–75)

Adjectives

#	Term (Adjective)	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
51	**Congruent**	**सर्वांगसम (Sarvaangsam)**	Identical in shape and size dimensions.	*"Prove these two triangles are congruent by the SAS rule."*
52	**Similar**	**समरूप (Samroop)**	Same shape with proportional corresponding sides.	*"Similar figures share equal angles but different side lengths."*
53	**Parallel**	**समांतर (Samantar)**	Coplanar lines maintaining constant separation.	*"Parallel lines have equal slopes and never intersect."*
54	**Perpendicular**	**लंबवत (Lambvat)**	Intersecting strictly at right angles (90°).	*"The normal vector is perpendicular to the tangent plane."*
55	**Continuous**	**संतत (Santat)**	Unbroken curve with no sudden jumps or holes.	*"Verify if the piecewise function is continuous at $x = 0$."*
56	**Discrete**	**असतत / विविक्त (Vivikt)**	Composed of distinct, separate countable values.	*"A Poisson variable models discrete random events."*
57	**Differentiable**	**अवकलनीय (Avkalaneey)**	Possessing a defined derivative everywhere in domain.	*"A sharp corner renders a curve non-differentiable."*
58	**Rational**	**परिमेय (Parimey)**	Expressible as a ratio of two whole integers.	*"Show that the repeating decimal represents a rational number."*
59	**Irrational**	**अपरिमेय (Aparimey)**	Non-terminating, non-repeating real number.	*"The square root of two is a classical irrational value."*
60	**Collinear**	**संरेखीय (Sanrekheey)**	Lying along one single common straight line.	*"Points A, B, and C are collinear because their slopes match."*
61	**Coplanar**	**समतलीय (Samtaleey)**	Occupying the exact same two-dimensional plane.	*"Three non-collinear points are always coplanar."*
62	**Equilateral**	**समबाहु (Sambaahu)**	Having all outer sides equal in length.	*"Each interior angle of an equilateral triangle measures 60°."*
63	**Isosceles**	**समद्विबाहु (Samdwibaahu)**	Possessing at least two equal sides and angles.	*"The base angles of an isosceles triangle are congruent."*
64	**Scalene**	**विषमबाहु (Vishambaahu)**	Triangle having three unequal side lengths.	*"In a scalene triangle, no two sides or angles match."*
65	**Orthogonal**	**लांबिक (Laambik)**	Perpendicular vectors with dot product zero.	*"Two non-zero vectors are orthogonal if their inner product is zero."*
66	**Asymmetric**	**असममित (Asammit)**	Lacking balance or invariant reflection symmetry.	*"The skewed distribution exhibits an asymmetric profile."*
67	**Symmetric**	**सममित (Sammit)**	Invariant under reflection or rotation transformation.	*"A normal bell curve is perfectly symmetric around its mean."*
68	**Homogeneous**	**समघाती (Samghaatee)**	Having terms of identical total algebraic degree.	*"Solve the first-order homogeneous differential equation."*
69	**Monotonic**	**एकदिष्ट (Ekdisht)**	Consistently non-increasing or non-decreasing.	*"The exponential function is strictly monotonic across all reals."*
70	**Asymptotic**	**अनंतस्पर्शीय (Anantsparsheey)**	Approaching a limiting boundary infinitely closely.	*"Examine the asymptotic behavior as x approaches infinity."*
71	**Finite**	**परिमित (Parimit)**	Bounded, measurable; having a countable limit.	*"This series contains a finite number of non-zero terms."*
72	**Infinite**	**अनंत / अपरिमित (Anant)**	Boundless, unlimited; larger than any real number.	*"The set of all prime numbers is provably infinite."*
73	**Singular**	**अव्युत्क्रमणीय (Avyutkramaneey)**	Square matrix having a determinant of zero.	*"A singular matrix cannot be inverted."*
74	**Convex**	**उत्तल (Uttal)**	Curving outward; internal segments stay inside.	*"All interior diagonals stay completely within a convex polygon."*
75	**Concave**	**अवतल (Avtal)**	Curving inward; having at least one reflex angle.	*"A concave polygon has at least one internal angle over 180°."*

PART 3: PRIMARY MATHEMATICAL VERBS (ITEMS 76–100)

Verbs: Action Terms

#	Term (Verb)	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
76	**Evaluate**	**मान ज्ञात करना (Maan Gyat Karna)**	Calculate numerical value of an expression.	*"Evaluate the algebraic expression when x equals negative three."*
77	**Differentiate**	**अवकलन करना (Avkalan Karna)**	Find derivative or rate of change function.	*"Differentiate the position vector with respect to time."*
78	**Integrate**	**समाकलन करना (Samakalan Karna)**	Find the antiderivative or cumulative area.	*"Integrate the piecewise curve between zero and two."*
79	**Simplify**	**सरल करना (Saral Karna)**	Reduce an expression to its most compact form.	*"Simplify the complex algebraic fraction by canceling common terms."*
80	**Factorize**	**गुणनखंड करना (Gunankhand Karna)**	Decompose a number/expression into products.	*"Factorize the quadratic expression into linear factors."*
81	**Substitute**	**प्रतिस्थापित करना (Pratisthapit)**	Replace variable with another expression/number.	*"Substitute the value of y back into the original equation."*
82	**Bisect**	**समद्विभाजित करना (Samdwibhajeet)**	Divide into two strictly congruent halves.	*"Construct an arc to bisect angle ABC accurately."*
83	**Extrapolate**	**बहिर्वेशन करना (Bahirveshan)**	Estimate trends beyond the known range.	*"Be cautious when you extrapolate data far past measured values."*
84	**Interpolate**	**अंतर्वेशन करना (Antarveshan)**	Estimate unknown values between known data points.	*"Interpolate linearly between five and six to find the midpoint."*
85	**Deduce**	**निष्कर्ष निकालना (Nishkarsh)**	Derive logical conclusion through valid reasoning.	*"From the established lemma, we can deduce the main result."*
86	**Approximate**	**सन्निकटन करना (Sannikatan)**	Estimate close to true value without exactness.	*"Approximate the irrational root to three decimal places."*
87	**Normalize**	**प्रसामान्यीकरण करना (Prasamanyikaran)**	Scale vector to unit length or probability to 1.	*"Normalize the eigenvector before formulating the matrix."*
88	**Eliminate**	**विलोपन करना (Vilopan Karna)**	Remove an unknown variable by algebraic operations.	*"Eliminate variable y by subtracting equation two from one."*
89	**Plot**	**आलेखित करना (Aalekhit Karna)**	Mark coordinates accurately onto a grid.	*"Plot the ordered pairs on the coordinate grid with labels."*
90	**Transpose**	**परिवर्त करना (Parivart Karna)**	Interchange rows and columns in a matrix.	*"Transpose matrix A and inspect whether it matches its negative."*
91	**Invert**	**व्युत्क्रम करना (Vyutkram Karna)**	Reverse operation, position, or find inverse.	*"To divide by a fraction, invert the divisor and multiply."*
92	**Converge**	**अभिसरित होना (Abhisarit Hona)**	Tend toward a finite limiting value.	*"Show that the infinite alternating series converges to ln 2."*
93	**Diverge**	**अपसरित होना (Apsarit Hona)**	Increase or oscillate without finite limit.	*"The harmonic series continues to diverge to positive infinity."*
94	**Maximize**	**अधिकतम करना (Adhiktam Karna)**	Find conditions producing highest possible output.	*"Find the dimensions that maximize the enclosed rectangular area."*
95	**Minimize**	**न्यूनतम करना (Nyoonatam Karna)**	Find conditions producing lowest possible value.	*"We seek to minimize the squared error between sample points."*
96	**Truncate**	**छोटा करना / काटना (Chhota Karna)**	Shorten by cutting off trailing digits.	*"Truncate the decimal at two digits without rounding."*
97	**Disprove**	**गलत सिद्ध करना (Galat Siddh)**	Demonstrate a conjecture is false by counterexample.	*"A single valid counterexample suffices to disprove the claim."*
98	**Subtend**	**अंतरित करना (Antarit Karna)**	Extend an arc or chord under a specific angle.	*"The arc subtends an angle of sixty degrees at the circle center."*
99	**Parametrize**	**प्राचलीकरण करना (Prachalikaran)**	Express curve variables in terms of a parameter.	*"Parametrize the circle boundary using cosine and sine."*
100	**Oscillate**	**दोलन करना (Dolan Karna)**	Fluctuate periodically between two fixed bounds.	*"The sine curve continues to oscillate between -1 and +1."*

PART 4: MATHEMATICAL ADVERBS (ITEMS 101–125)

Adverbs

#	Term (Adverb)	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
101	**Monotonically**	एकदिष्ट रूप से (Ekdisht Roop Se)	Moving continually in one direction without reversing.	*"The sequence increases monotonically toward the upper bound."*
102	**Asymptotically**	अनंतस्पर्शी रूप से (Anantsparsheey)	Approaching a limiting state as parameter nears infinity.	*"The two functions behave asymptotically identically for large n."*
103	**Conversely**	इसके विपरीत / विलोमतः (Vilomtah)	Introducing the reverse of a previous statement.	*"If a figure is a square, it is a rectangle; conversely, not every rectangle is a square."*
104	**Respectively**	क्रमशः (Kramashah)	In the identical sequential order already mentioned.	*"The roots of the system are 2, 4, and 8, respectively."*
105	**Strictly**	सख्ती से / पूर्णतः (Sakhti Se)	Excluding equality or loose boundary conditions.	*"The function is strictly greater than zero across its domain."*
106	**Identically**	सर्वसम रूप से (Sarvasam Roop Se)	Equaling for every conceivable choice of input variable.	*"The expression is identically zero for all real values of x."*
107	**Approximately**	लगभग (Lagbhag)	Very close to, but not strictly exact in precision.	*"The ratio is approximately equal to 3.1416."*
108	**Inversely**	व्युत्क्रमानुपाती रूप से (Vyutkram)	Varying such that one increases as other decreases.	*"Gravitational attraction is inversely proportional to distance squared."*
109	**Directly**	सीधे / अनुक्रमानुपाती (Seedhe)	Varying together at a constant multiplicative ratio.	*"Displacement varies directly with elapsed travel time."*
110	**Uniquely**	अद्वितीय रूप से (Adwiteey Roop Se)	Existing in exactly one distinct manner.	*"Every integer factors uniquely into a product of primes."*
111	**Linearly**	रैखिक रूप से (Raikhik Roop Se)	Changing at a constant, straight-line rate.	*"Cost scales linearly as the quantity produced expands."*
112	**Exponentially**	चरघातांकी रूप से (Charghaataanki)	Compounding at rates proportional to current value.	*"Bacterial population grows exponentially under ideal nutrient levels."*
113	**Arbitrarily**	स्वेच्छया / मनमाने ढंग से (Swecchya)	Selected without preset constraint or bias.	*"Choose an arbitrarily small epsilon greater than zero."*
114	**Symmetrically**	सममित रूप से (Sammit Roop Se)	Evenly distributed across a central reflection line.	*"The data points scatter symmetrically about the regression line."*
115	**Continuously**	लगातार / निरंतर (Nirantar)	Proceeding smoothly without interruption or break.	*"Interest compounds continuously under Euler's constant."*
116	**Mutually**	परस्पर (Paraspar)	Shared reciprocally between two or more items.	*"Two events are mutually exclusive if they cannot happen together."*
117	**Inductively**	आगमनात्मक रूप से (Aagmanaatmak)	Proved via mathematical induction steps.	*"We prove inductively that the formula holds for all natural n."*
118	**Deductively**	निगमनात्मक रूप से (Nigmanaatmak)	Derived necessarily from established axiomatic premises.	*"The corollary follows deductively from the previous theorem."*
119	**Empirically**	प्रयोगसिद्ध रूप से (Prayogsiddh)	Determined through observational statistical measurement.	*"The probability was estimated empirically through ten thousand trials."*
120	**Equivalently**	समतुल्य रूप से (Samtulya Roop Se)	Stating an interchangeable mathematical form.	*"Equivalently, one can multiply by the inverse matrix."*
121	**Pointwise**	बिंदुवार (Bindu-vaar)	Evaluated individually at each individual point.	*"The sequence converges pointwise, though not uniformly."*
122	**Uniformly**	एकसमान रूप से (Eksamaan Roop Se)	Holding identically throughout the whole domain.	*"The approximation converges uniformly on closed intervals."*
123	**Periodically**	आवर्ती रूप से (Aavarti Roop Se)	Repeating exact pattern values at set regular intervals.	*"The wave oscillates periodically with a period of 2π."*
124	**Tangentially**	स्पर्शरेखीय रूप से (Sparshrekheey)	Touching grazing boundaries without cutting across.	*"The sphere contacts the containment plane tangentially."*
12[illegible]	**[illegible]**	[illegible]	Occurring or satisfying conditions at [illegible]	*"Find the coordinates that solve both linear [illegible]"*

PART 5: PRONOUNS, CONJUNCTIONS & INTERJECTIONS (ITEMS 126–150)

Structural Connectors

#	Grammar Category	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
126	Each (Pronoun)	प्रत्येक (Pratyek)	Every member considered individually in a set.	*"Each of these triangles shares a common circumcircle."*
127	Neither (Pronoun)	दोनों में से कोई नहीं (Dono Mein Se Koi Nahi)	Not one nor the other of two items.	*"Neither candidate root satisfies the boundary condition."*
128	Either (Pronoun)	दोनों में से एक (Dono Mein Se Ek)	One or the other of two alternatives.	*"Either test statistic can validate our alternative hypothesis."*
129	Which (Pronoun)	जो / कौन सा (Jo / Kaun Sa)	Relative pronoun specifying a clause or set item.	*"Substitute x = 4, which eliminates our square root term."*
130	Whatever (Pronoun)	जो कुछ भी (Jo Kuchh Bhi)	Anything regardless of value or selection.	*"Whatever value of k you choose, the determinant remains fixed."*
131	None (Pronoun)	कोई नहीं (Koi Nahi)	Not any of a given collection or subset.	*"None of the proposed coordinates lie within the third quadrant."*
132	Both (Pronoun)	दोनों (Dono)	The two items considered collectively together.	*"Both curves intersect at the origin."*
133	All (Pronoun)	सभी (Sabhi)	The entire aggregate quantity without exception.	*"All prime numbers greater than two are necessarily odd."*
134	Whichever (Pronoun)	जो भी (Jo Bhi)	Any item chosen without differential outcome.	*"Whichever branch you differentiate, the gradient evaluates equally."*
135	Any (Pronoun)	कोई भी (Koi Bhi)	An arbitrary element from an unspecified set.	*"Is there any non-trivial solution to this matrix system?"*
136	Therefore (Conj.)	अतः / इसलिए (Atah / Isliye)	For that logical reason; introducing a deduction.	*"The determinant is zero; therefore, the matrix is singular."*
137	Hence (Conj.)	अतएव / फलस्वरूप (Ataev)	As a consequence; drawing direct formal result.	*"Sides match in ratio; hence, the two triangles are similar."*
138	Thus (Conj.)	इस प्रकार (Is Prakaar)	In this specified manner; indicating conclusion.	*"Both derivatives agree; thus, the function remains smooth."*
139	Provided that (Conj.)	बशर्ते कि (Basharte Ki)	On the explicit conditional requirement that.	*"The identity holds provided that x does not equal unity."*
140	Since / As (Conj.)	चूंकि (Chunki)	Because; presenting known or proven fact.	*"Since the slope is negative, the curve decreases monotonically."*
141	Unless (Conj.)	जब तक कि... न (Jab Tak Ki... Na)	Except on the condition that.	*"You cannot cancel variables unless you verify they are non-zero."*
142	Whereas (Conj.)	जबकि (Jabki)	In contrast or comparison with the fact that.	*"Scalars have size, whereas vectors possess directed magnitude."*
143	Whenever (Conj.)	जब कभी भी (Jab Kabhi Bhi)	At any time or under every condition when.	*"Whenever sine reaches zero, the tangent graph crosses the axis."*
144	Albeit (Conj.)	यद्यपि (Yadyapi)	Even though; acknowledging a qualification.	*"It produces a convergent, albeit exceptionally slow, solution."*
145	Lest (Conj.)	ऐसा न हो कि (Aisa Na Ho Ki)	To avoid the unintended consequence of.	*"Enclose expressions in brackets lest signs be misapplied."*
146	Recall! (Interj.)	याद रखें! (Yaad Rakhein!)	Classroom prompt urging students to recall an earlier rule.	*"Recall! The derivative of cosine carries a negative sign."*
147	Notice! (Interj.)	ध्यान दीजिए! (Dhyan Deejiye!)	Focus marker directing student gaze to critical detail.	*"Notice! The entire middle term cancels out completely here."*
148	Beware! (Interj.)	सावधान! (Saavdhan!)	Pedagogical alert cautioning against common errors.	*"Beware! Do not divide both sides by zero."*
149	Look! (Interj.)	देखिए! (Dekhiye!)	Prompt highlighting an emerging visual or pattern.	*"Look! The graph exhibits symmetry about the y-axis."*
150	Bingo! (Interj.)	अरे वाह! / बिल्कुल सही! (Shabash)	Spoken celebratory cue upon finding the exact solution.	*"Substitute five, and bingo! The identity balances."*

PART 6: SPOKEN MATHEMATICAL PHRASAL VERBS (ITEMS 151–175)

Phrasal Verbs: Set A

#	Phrasal Verb	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
151	Cancel out	एक-दूसरे को काटना (Kaatna)	Neutralize or eliminate terms of equal opposite value.	*"The positive and negative sine components cancel out."*
152	Factor out	उभयनिष्ठ बाहर निकालना (Bahar Nikalna)	Extract a common divisor outside parentheses.	*"Factor out four from each term before taking the radical."*
153	Plug in	मान रखना (Maan Rakhna)	Insert an explicit numerical value into an expression.	*"Plug in x = 2 to discover the corresponding y coordinate."*
154	Rule out	खारिज करना (Khaarij Karna)	Dismiss impossible or extraneous solutions.	*"We can rule out negative lengths since distance is positive."*
155	Work out	हल निकालना (Hal Nikalna)	Calculate stepwise progression to arrive at an answer.	*"Take two minutes to work out the remainder on paper."*
156	Cross out	काट देना (Kaat Dena)	Draw a line through matched canceled expressions.	*"Cross out matching binomial factors across the fraction bar."*
157	Break down	विभाजित करना (Vibhajeet Karna)	Decompose a complex multi-step problem into steps.	*"Let us break down this composite solid into a cone and cylinder."*
158	Add up	योग करना (Yog Karna)	Sum individual scalar numbers together.	*"Add up the interior angles to ensure they total 180°."*
159	Count down	उल्टी गिनती करना (Ulti Ginti)	Recite values in reverse descending order.	*"Count down from ten by steps of one point five."*
160	Round off	सन्निकट करना (Sannikat Karna)	Adjust to nearest whole integer or decimal mark.	*"Round off the final decimal quotient to two decimal places."*
161	Narrow down	सीमित करना (Seemit Karna)	Reduce possible search bounds to fewer candidates.	*"Bisection lets us narrow down the interval containing the root."*
162	Scale up	आकार बढ़ाना (Aakaar Badhana)	Enlarge proportions by a fixed dilation scalar.	*"Scale up the model triangle by a magnification factor of three."*
163	Scale down	आकार घटाना (Aakaar Ghatana)	Shrink proportions uniformly across all dimensions.	*"Scale down the coordinates by dividing throughout by ten."*
164	Map onto	प्रतिचित्रित करना (Pratichitrit)	Transform every point of domain to target range.	*"The reflection matrix maps point P onto P-prime directly."*
165	Sum up	संक्षेप करना / योग करना (Sankshep)	Tally series elements or synthesize the lesson theorem.	*"To sum up, quadratic equations always yield two complex roots."*
166	Take away	घटाना (Ghataana)	Subtract a specified quantity from another.	*"If you take away seven from fifteen, eight remains."*
167	Write down	लिख लेना (Likh Lena)	Record formula or data carefully on paper/board.	*"Write down the standard quadratic formula in your notebook."*
168	Set up	स्थापित करना (Sthapit Karna)	Construct an algebraic model or geometric coordinate grid.	*"Set up the definite integral before choosing substitution rules."*
169	Level off	स्थिर हो जाना (Sthir Ho Jaana)	Flatten curve slope toward a horizontal asymptote.	*"The growth velocity levels off as carrying capacity nears."*
170	Even out	संतुलित होना (Santulit Hona)	Attain uniformity; smooth fluctuating measurements.	*"Sample fluctuations even out over large iteration counts."*
171	Zero in on	केंद्रित होना (Kendrit Hona)	Focus progressively onto the target exact coordinate.	*"Newton's method zeros in on the real root with quadratic speed."*
172	Drop out	लुप्त हो जाना (Lupt Ho Jaana)	Disappear from equation due to multiplication by zero.	*"Higher order terms drop out when evaluating at zero."*
173	Carry over	हासिल आगे ले जाना (Haasil Le Jaana)	Transfer tens digit to the adjacent column during addition.	*"Add six and seven to get thirteen; write three, carry over one."*
174	Point out	संकेत करना (Sanket Karna)	Draw student attention to a significant relationship.	*"Let me point out why this base step validates the proof."*
175	Boil down to	सार तत्व होना (Saar Tatva Hona)	Reduce in essence to a fundamental relationship.	*"This proof boils down to demonstrating the positivity of discriminant."*

PART 7: PHRASAL VERBS & CLASSROOM PHRASES (ITEMS 176–200)

Phrasal Verbs & Instructional Phrases

#	Phrasal Verb / Phrase	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
176	**Divide through**	**पूरी समीकरण को भाग देना**	Divide every term of an entire equation by a common value.	*"Divide through by the leading coefficient to make it monic."*
177	**Multiply out**	**गुणा करके खोलना (Guna Karna)**	Expand bracketed algebraic products.	*"Multiply out both binomial factors using the FOIL method."*
178	**Tidy up**	**सुव्यवस्थित करना (Suvyavasthit)**	Group terms and simplify an algebraic presentation.	*"Tidy up your working by collecting all common powers."*
179	**Cross-multiply**	**वज्र-गुणन करना (Vajra-Gunan)**	Multiply diagonal numerator-denominator across equals sign.	*"Cross-multiply the equality of fractions to clear denominators."*
180	**Bring down**	**नीचे उतारना (Neeche Utaarna)**	Lower the next sequential digit in long division.	*"Bring down the next digit from the dividend to continue division."*
181	**Step by step**	**क्रमशः / चरणबद्ध (Charanbaddh)**	Proceeding deliberately through each operational phase.	*"Solve this differential equation step by step on the board."*
182	**Without loss of generality**	**व्यापकता की हानि के बिना**	Assuming a specific case that mirrors all other cases identically.	*"Without loss of generality, assume variable a exceeds b."*
183	**If and only if (iff)**	**यदि और केवल यदि**	Denoting bidirectional logical equivalence.	*"An integer is even if and only if its last digit is divisible by 2."*
184	**Q.E.D.**	**इति सिद्धम् (Iti Siddham)**	Notation marking conclusion of a formal proof.	*"Both sides match identically; hence, Q.E.D."*
185	**Under the condition that**	**इस शर्त के अधीन कि**	Subject strictly to specified restrictions.	*"The identity holds under the condition that denominator is non-zero."*
186	**In terms of**	**के पदों में (Ke Padon Mein)**	Expressed solely with a designated variable.	*"Express the volume of the sphere strictly in terms of radius."*
187	**By inspection**	**निरीक्षण द्वारा (Nireekshan Dwara)**	Determined instantly through immediate observation.	*"By inspection, we see that x = 1 satisfies this cubic system."*
188	**By contradiction**	**विरोधाभास द्वारा (Virodhabhas)**	Proving a claim by showing its negation is absurd.	*"We prove the irrationality of radical two by contradiction."*
189	**Vice versa**	**इसके विपरीत भी (Iske Vipreet Bhi)**	The reverse proposition is equally valid.	*"Every equilateral triangle is equiangular, and vice versa."*
190	**On the contrary**	**इसके विपरीत (Iske Vipreet)**	Introducing an assertion opposing the prior statement.	*"We assumed p was composite; on the contrary, it has no factors."*
191	**Up to a constant**	**एक नियतांक तक (Niyataank Tak)**	Identical except for an added arbitrary scalar.	*"Indefinite integration determines the primitive up to a constant."*
192	**To the nearest...**	**निकटतम... तक (Nikat-tam)**	Approximated to specified precision unit.	*"Compute the answer to the nearest thousandth."*
193	**Collect like terms**	**समान पदों को एकत्र करना**	Group matching powers and variables together.	*"Collect like terms before moving variables to the left side."*
194	**Clear the fractions**	**भिन्नों को हटाना (Bhinnon Ko)**	Multiply through by least common denominator.	*"Clear the fractions by multiplying every single term by 12."*
195	**Equate to zero**	**शून्य के बराबर रखना (Shunya)**	Set expression equal to zero to discover stationary roots.	*"Equate the first derivative to zero to locate critical points."*
196	**As a matter of course**	**स्वाभाविक रूप से (Swabhavik)**	Proceeding naturally from established methodology.	*"Check domain restrictions as a matter of course."*
197	**In reverse order**	**उलटे क्रम में (Ulte Kram Mein)**	Arranged backward from finish to origin.	*"Write the polynomial terms in reverse order of powers."*
198	**Take the reciprocal**	**व्युत्क्रम लेना (Vyutkram Lena)**	Invert fraction so numerator and denominator trade.	*"Take the reciprocal of both sides to isolate the variable."*
199	**Hold constant**	**स्थिर रखना (Sthir Rakhna)**	Maintain one variable fixed during partial differentiation.	*"Differentiate with respect to x while holding y constant."*
200	**To scale**	**पैमाने के अनुसार (Paimane Ke)**	Drawn in exact proportion to actual dimensions.	*"Note that this illustrative triangle is not drawn to scale."*

PART 8: ACADEMIC MATHEMATICAL PHRASES (ITEMS 201–225)

Academic Phrases

#	Mathematical Phrase	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
201	**Subject to**	**के अधीन / की शर्तों पर**	Constrained by an auxiliary operational condition.	*"Maximize profit subject to the budget constraint."*
202	**With respect to**	**के सापेक्ष (Ke Saapeksh)**	Indicating reference variable of differentiation/change.	*"Differentiate volume with respect to time to get flow rate."*
203	**In the limit**	**सीमांत स्थिति में (Seemant)**	As a designated variable approaches its boundary limit.	*"In the limit as delta x approaches zero, we find slope."*
204	**Well-defined**	**सुपरिभाषित (Suparibhashit)**	Unambiguous, consistent, and mathematically sound.	*"A function must assign a well-defined value to each input."*
205	**Non-trivial**	**अतुच्छ / सार्थक (Atuchh)**	Meaningful; excluding the obvious zero solution.	*"The homogeneous system yields non-trivial solutions if det(A) = 0."*
206	**Order of magnitude**	**परिमाण की कोटि (Parimaan Koti)**	Class of scale based on powers of ten.	*"The calculated radius is two orders of magnitude smaller."*
207	**Degrees of freedom**	**स्वातंत्र्य कोटि (Swaatantrya)**	Number of independently variable parameters in system.	*"A sample of size fifteen has fourteen degrees of freedom."*
208	**In accordance with**	**के अनुसार (Ke Anusaar)**	Complying rigorously with established axioms.	*"Construct the perpendicular bisector in accordance with rules."*
209	**It follows that**	**इससे यह निष्कर्ष निकलता है**	Indicating immediate valid deduction from prior step.	*"Both slopes are equal; it follows that lines are parallel."*
210	**Sufficient condition**	**पर्याप्त शर्त (Paryapt Shart)**	Condition that guarantees the truth of statement.	*"Differentiability is a sufficient condition for continuity."*
211	**Necessary condition**	**आवश्यक शर्त (Aavashyak Shart)**	Condition that must hold for statement to be true.	*"Continuity is a necessary condition for differentiability."*
212	**Mutually exclusive**	**परस्पर अपवर्जी (Paraspar Apvarji)**	Events that cannot occur at the same instant.	*"Getting heads and tails on a single coin toss are mutually exclusive."*
213	**Independent of**	**से स्वतंत्र (Se Swatantra)**	Unaffected by changes in another parameter.	*"The circle's radius is independent of coordinate translations."*
214	**In closed form**	**संवृत रूप में (Samvrit Roop)**	Expressed using finite standard functions.	*"The integral cannot be evaluated in closed form."*
215	**To begin with**	**शुरू करने के लिए (Shuru Me)**	Opening foundational phase of mathematical proof.	*"To begin with, define your universe of discourse clearly."*
216	**In other words**	**दूसरे शब्दों में (Doosre Shabdon)**	Rephrasing for conceptual classroom clarity.	*"In other words, the function possesses no local extrema."*
217	**At first glance**	**पहली नज़र में (Pehli Nazar)**	Appearing obvious prior to rigorous testing.	*"At first glance the series looks divergent, but it converges."*
218	**By virtue of**	**के आधार पर / के कारण**	By authority or logic of a proven identity.	*"By virtue of symmetry, the integral from -a to a evaluates to zero."*
219	**On the grounds that**	**इस आधार पर कि (Is Aadhar Par)**	Presenting mathematical justification for step.	*"We reject the negative root on the grounds that time is positive."*
220	**In a similar vein**	**इसी प्रकार से (Isi Prakaar Se)**	Proceeding using an identical method of proof.	*"In a similar vein, one can compute the y-intercept."*
221	**For the sake of**	**के उद्देश्य से (Ke Uddeshya Se)**	For convenience, clarity, or simplification.	*"For the sake of brevity, let us denote the matrix as M."*
222	**As opposed to**	**इसके विपरीत (Iske Vipreet)**	Contrasting two different concepts.	*"We use an open interval as opposed to a closed boundary."*
223	**In direct contrast**	**सीधे विपरीत (Seedhe Vipreet)**	Highlighting stark oppositional property.	*"In direct contrast to rational numbers, pi never repeats."*
224	**From this viewpoint**	**इस दृष्टिकोण से (Is Drishtikon)**	Interpreting a relation from a geometric perspective.	*"From this viewpoint, differentiation finds line inclination."*
225	**All other things being equal**	**अन्य बातें समान रहने पर**	Holding outside factors unchanging.	*"All other things being equal, increasing variance widens the curve."*

PART 9: MATHEMATICAL IDIOMS & COLLOQUIALISMS (ITEMS 226–250)

Idioms & Expressions

#	Idiom / Saying	Hindi Meaning	English Pedagogical Meaning	Spoken Classroom / Writing Example
226	**It doesn't add up**	**बात समझ में न आना / मेल न खाना**	A conclusion or solution is illogical or inconsistent.	*"Check your balance; these arithmetic figures simply don't add up."*
227	**Back to square one**	**शुरुआती बिंदु पर लौटना**	Forced to restart from scratch after reaching an impasse.	*"The identity failed; we are back to square one with our proof."*
228	**Do a 180 (turnaround)**	**पूरी तरह बदल जाना (180° मोड़)**	Completely reverse opinion, sign, or trajectory.	*"Multiplying by a negative causes the inequality to do a 180."*
229	**Go off on a tangent**	**मुद्दे से भटक जाना**	Diverge abruptly into an unrelated peripheral topic.	*"Focus on factoring; do not go off on a tangent about history."*
230	**Square the circle**	**असंभव कार्य करना**	Attempting an intrinsically impossible task.	*"Solving this insolvable quintic algebraically is like squaring the circle."*
231	**Vicious circle**	**दुष्चक्र (Dushchakra)**	Circular feedback loop of errors reinforcing one another.	*"Circular definitions trap your mathematical logic in a vicious circle."*
232	**Draw the line**	**सीमा तय करना (Seema Tay Karna)**	Set a rigid boundary or tolerance threshold.	*"We draw the line at three decimal points of numerical rounding error."*
233	**Even the score**	**बराबर करना (Barabar Karna)**	Restore balance or parity across sides.	*"Subtract twelve from both sides to even the algebraic score."*
234	**The lesser of two evils**	**दो बुराइयों में से छोटी बुराई**	Selecting the simpler of two tedious calculation paths.	*"Integration by parts here is the lesser of two evils compared to trig substitution."*
235	**A fraction of**	**एक छोटा सा हिस्सा (Chhota Hissa)**	A very small portion or percentage of a total.	*"Computing via matrix shortcuts took a fraction of the usual time."*
236	**Ballpark figure**	**मोटा-मोटा अनुमान (Anumaan)**	A rough, acceptable order-of-magnitude estimate.	*"Before exact integration, give me a ballpark figure for the area."*
237	**At odds with**	**के विपरीत होना (Ke Vipreet)**	In direct contradiction with known theorems.	*"This calculated negative variance is at odds with probability laws."*
238	**Rule of thumb**	**अंगूठा नियम / व्यावहारिक नियम**	Practical, easily applied heuristic principle.	*"As a rule of thumb, check roots whenever you divide by an unknown."*
239	**Crunch numbers**	**भारी गणनाएं करना (Gannayein)**	Perform extensive, repetitive numerical arithmetic.	*"Let our computational software crunch the numbers for large n."*
240	**In full swing**	**पूरे जोर-शोर से (Poore Zor-Shor)**	Operating at the highest level of active progress.	*"Our theorem revision session is now in full swing."*
241	**On the same wavelength**	**एक ही विचार पर होना**	Sharing identical understanding and clarity.	*"Good! The whole class is on the same wavelength regarding limits."*
242	**By the book**	**नियमानुसार (Niyamanusaar)**	Strictly adhering to formal standardized protocols.	*"Execute the Gaussian elimination strictly by the book."*
243	**Tip of the iceberg**	**समस्या का छोटा सा दिखने वाला हिस्सा**	Small observable part of a much deeper concept.	*"Single-variable calculus is just the tip of the iceberg in analysis."*
244	**Bottom line**	**मुख्य बात / निष्कर्ष (Mukhya Baat)**	The fundamental, decisive final fact.	*"The bottom line is that any division by zero is undefined."*
245	**Nip in the bud**	**शुरुआत में ही खत्म करना**	Correct a conceptual misconception before it compounds.	*"Nip that sign mistake in the bud before expanding parentheses."*
246	**Play by the rules**	**नियमों का पालन करना**	Adhere strictly to axioms and algebraic laws.	*"You cannot cancel across addition; you must play by the rules."*
247	**Make headway**	**प्रगति करना (Pragati Karna)**	Make measurable progress in solving a complex problem.	*"We are making solid headway on this multi-variable optimization."*
248	**Grain of salt**	**संदेह के साथ स्वीकार करना**	Accepting experimental estimates with caution.	*"Take the extrapolated statistical result with a grain of salt."*
249	**See eye to eye**	**पूरी तरह सहमत होना (Sehmat Hona)**	Agreeing completely on a result or method.	*"Both analytical and numerical methods see eye to eye on this root."*
250	**Zero-sum game**	**शून्य-योग खेल (Shunya-Yog Khel)**	Situation where one party's gain equals the other's loss.	*"In game theory, poker constitutes a classical zero-sum game."*